

AZ-305 Azure Solutions Architect — Cram Guide

Designing Microsoft Azure Infrastructure Solutions

Version: 2.0 — Migration Heavy Edition | 2026

■ Exam at a Glance

Prerequisite	AZ-104 Azure Administrator Associate
Questions	Case-study based (not trivia)
Duration	180 minutes
Passing Score	700 / 1000
Cost	~\$165 USD
Renewal	Annual — free assessment on Microsoft Learn

Domain Weightings (2026)

Domain	Weight	Key Topics
Design Identity & Governance	25–30%	Entra ID, RBAC, PIM, Policy, Blueprints, Landing Zones
Design Data Storage	25–30%	Storage accounts, Azure Files, Blob, SQL, Cosmos DB, Data Lake
Design Infrastructure	25–30%	VMs, Containers, AKS, App Service, Functions, HA/DR
Design Migration	5–10%	Azure Migrate, DMS, ASR, HCX, assessment patterns
Design Business Continuity	10–15%	Backup, ASR, availability zones, failover

****Three domains carry 85–90% of the exam.** Master Identity/Governance, Data Storage, and Infrastructure first.**

■ DOMAIN 1 — Identity, Governance & Monitoring (25–30%)

Microsoft Entra ID (formerly Azure AD)

What it is: Cloud-based identity and access management. Handles authentication for all Azure resources, SaaS apps, and custom LOB applications.

Tiers & Plans

Feature	Free	P1	P2
Users	500k	Unlimited	Unlimited
SSO (federated)	10 apps	Unlimited	Unlimited
MFA	■	■	■
Conditional Access	■	■	■
Identity Protection	■	■	■
Access Reviews	■	■	■
Entitlement Management	■	■	■
P2: \$6/user/mo			

User Identities: Cloud vs. Synchronized vs. Federated

Cloud-only identities:

- Pure Azure AD — no on-prem footprint

- Best for: fully cloud-native orgs, new tenants
- Passwords stored hash in Azure AD

Password Hash Sync (PHS):

- Azure AD Connect syncs password hashes from on-prem AD
- Enables seamless SSO + cloud auth fallback
- Simple to deploy, low complexity
- Best for: orgs wanting hybrid but avoiding full ADFS

Pass-Through Authentication (PTA):

- Agent on-prem validates credentials directly against your AD
- No password hashes stored in cloud
- Best for: high security requirements, no password hash storage in cloud

Federation (ADFS / Ping / etc.):

- Full trust chain between on-prem IdP and Entra ID
- Supports complex auth policies, smart card, third-party MFA
- Highest complexity — requires on-prem infrastructure uptime
- Best for: regulatory requirements, existing ADFS investment

Hybrid Identity — Azure AD Connect

Azure AD Connect: The bridge between on-prem AD and Entra ID.

- **Express Settings:** Single forest, password hash sync, no custom config
- **Custom Settings:** Multi-forest, multiple sync engines, Exchange hybrid, LDAP v3
- **Staging Mode:** Test config without affecting production sync
- **Azure AD Connect Health:** Monitor sync errors, agent health, sign-in activity

Multi-forest scenarios:

- Use Azure AD Connect with multiple connectors
- Separate sync servers per forest feeding into single Azure AD tenant
- Consider MIM (Microsoft Identity Manager) for complex identity consolidation

Application Identities — Service Principals vs. Managed Identities

Service Principal:

- App registration in Entra ID
- Client secret or certificate for auth
- You manage the credential lifecycle (rotation, expiry)
- Used for: DevOps, automation, CI/CD pipelines

System-assigned Managed Identity:

- Azure manages the credential
- Auto-rotation of secrets
- Single Azure resource (VM, App Service, Function, etc.)
- No credential management overhead
- Preferred for: Azure-hosted workloads

User-assigned Managed Identity:

- Standalone Azure resource
- Can be shared across multiple Azure resources
- Credential lifecycle managed by you
- Best for: multiple VMs / services sharing one identity

Conditional Access — The Key Concepts

Conditional Access = **If this** → **Then that** for access decisions.

Common signals:

- User risk (compromised account?)
- Sign-in risk (anomalous login?)
- Device compliance (MDM enrolled? healthy?)
- IP location (trusted locations?)
- Application (which SaaS/app?)

Grant controls:

- Require MFA
- Require compliant device

- Require approved client app
- Require app protection policy
- Block access

Session controls:

- Sign-in frequency
- Persistent browser session
- Customize app token lifetime

Exam pattern: "Users are being prompted for MFA when accessing from unfamiliar locations" → Configure named locations + conditional access policy.

Privileged Identity Management (PIM)

PIM = Just-in-time elevated access. No standing privileges.

Key concepts:

- **Eligible assignments:** Pre-approved roles you can activate for a time window
- **Activation:** You request, approver grants, you get role for max duration
- **Time-bound:** Roles expire automatically
- **Audit trail:** Every activation is logged
- **Justification required:** Document why you need the role

Who needs it: Global Admin, Owner, Contributor on production subscriptions, security roles.

MFA enforcement in PIM: Always enforced at activation — even if tenant MFA is off.

Access Reviews

Periodic re-certification campaigns:

- "Manager confirms: does this user still need this role?"
- Self-review option
- Auto-remediation: remove access if no response
- Use for: guest accounts, temporary project access, role creep

Azure Policy & Blueprints

Azure Policy

Policy = Enforce compliance at scale. Evaluates *existing* and *new* resources.

Key built-in initiatives:

- Audit Windows VMs → Check for encryption, endpoints
- Enable Monitoring in Azure Security Center → ASC recommendations
- NIST SP 800-53 → Regulatory compliance
- ISO 27001 → ISO certification alignment

Policy vs. RBAC:

	Policy	RBAC
Scope	Resources (what)	Who can do what
Enforces	Configuration/compliance	Access rights
Evaluates	Existing + new resources	At access time
Assignment	Management groups, subscriptions, RGs	Users, groups, service principals
Effect	deny, audit, append, modify, deployIfNotExists	allow, deny

Policy effects (know for exam):

- deny — Blocks resource creation
- audit — Logs non-compliance, doesn't block
- deployIfNotExists — Deploys a template if condition met
- append — Adds fields to existing resources
- modify — Updates existing resource tags or properties

Exemptions: You can exempt a scope from a policy for a time period or indefinitely — useful for break-glass scenarios.

Azure Blueprints

Blueprint = Package of artifacts (policies, ARM templates, role assignments) that you can assign to *new* subscriptions repeatedly.

Artifacts in a blueprint:

- Resource groups
- ARM templates (deploy storage, SQL, etc.)
- Policy assignments
- Role assignments
- ARM template specs

Assignment vs. definition:

- Define once in a management group
- Assign to subscriptions — each subscription gets identical baseline

Difference from Policy:

- Policy evaluates existing resources
- Blueprint sets up *new* environments with standard baseline
- Blueprint is subscription lifecycle (applies at creation)

Azure Landing Zones (ALZ)

What is a Landing Zone?

A landing zone is the **environment foundation** — pre-configured, scalable, secure. It's where your workloads live.

Core principles:

- Policy-driven governance from day one
- Network segmentation (hub-and-spoke or Virtual WAN)
- Centralized identity, security, and monitoring
- Subscription vending (on-demand provisioning)

Management Group Hierarchy

```
Tenant Root Group
■■■ Sandbox (never deploy here)
■■■ Corp (production workloads)
■ ■■■ Prod (production)
■ ■ ■■■ Prod-EastUS
■ ■ ■■■ Prod-WestEurope
■ ■■■ NonProd (dev/test)
■ ■■■ Dev
■ ■■■ Test
■■■ Online (public-facing: CDN, Front Door)
■■■ Decommissioned
```

Key principles:

- Top-level management group: org name
- Don't put subscriptions directly under root
- Corp + Online separation: corp = private connectivity, online = public internet
- Don't grant access at root — manage at management group level

Hub-and-Spoke Architecture

[illegible]

Hub components:

- Azure Firewall (or third-party NVA) — all traffic inspection
- VPN Gateway or ExpressRoute — on-prem connectivity
- Azure Bastion — jump box replacement
- Private DNS zones — centralized DNS resolution
- VPN Gateway or ER for cross-region redundancy

When to use hub-spoke:

- Standard enterprise topology
- Need centralized security inspection
- Multiple workloads need shared services
- Requires network segmentation

When NOT to use:

- Very simple single-app scenarios
- Global distributed orgs → consider Virtual WAN instead

Azure Virtual WAN

- Fully managed Microsoft-managed SD-WAN
- Hub redundancy built-in
- Global transit network
- Good for: large multi-region orgs, many branch offices
- Replaces traditional hub-spoke for global topology

Azure Firewall Premium

Feature	Standard	Premium
Stateful firewall	■	■
Threat intelligence	■	■
FQDN filtering	■	■
TLS inspection	■	■
IDPS	■	■
Web categories	■	■
Malware inspection	■	■

When Premium: If you need to inspect outbound TLS traffic, need IDPS, or URL filtering with categories.

Monitoring — Azure Monitor + Sentinel**Azure Monitor — The Foundation****Data sources:**

- Application logs (custom app telemetry)
- VM agents (Log Analytics)
- Azure resource metrics (built-in)
- Activity logs (subscription-level)
- Container insights (AKS)
- Network monitoring (NSG flow logs, VPN diagnostics)

Log Analytics:

- Workspace-based architecture
- Multiple subs can send to one workspace (or separate per workload)
- Data retention: 30–730 days (pay per GB after free tier)
- Use Azure Monitor private link scope for private connectivity

Kusto Query Language (KQL): You need basic KQL for the exam.

```
// Top 10 failing requests
requests
| where timestamp > ago(1d)
| where success == false
| summarize count() by name, resultCode
| order by count_desc
| take 10
```

Alerts — Know the Types

Alert Type	When to Use
Metric alerts	Threshold breach on CPU, memory, latency
Log alerts	Complex KQL queries on log data
Activity log alerts	Azure resource lifecycle events
Smart groups	AI groups related alerts to reduce noise
Prometheus (Container Apps)	Metrics for AKS

Action groups: Who gets notified and how (email, SMS, webhook, ITSM, automation runbook)

Azure Sentinel — SIEM + SOAR

- Cloud-native SIEM (Security Information and Event Management)
- Collects from multiple sources: Entra ID, Office 365, Azure resources, AWS, GCP
- Uses ML/AI to detect threats (Fusion detection)
- **Incidents** = correlated alerts grouped by attack story
- **Playbooks** = automated SOAR responses (Logic Apps)
- **MITRE ATT&CK; matrix** — map detected activity to tactics

When you need Sentinel:

- Enterprise-wide security monitoring
- Need to correlate across multiple cloud platforms
- Compliance requires SIEM
- Need automated incident response

■ DOMAIN 2 — Data Storage Solutions (25–30%)

Azure Storage Accounts

Every storage account = one resource with multiple data services.

SKU Comparison

SKU	Redundancy Options	Use When
Standard General Purpose v2	LRS, ZRS, GRS, GZRS	Most workloads — blobs, files, tables, queues
Premium Block Blobs	LRS, ZRS	High throughput, low latency — log files, media
Premium File Shares	LRS, ZRS	Enterprise file shares, SMB, lift-and-shift
Premium Page Blobs	LRS only	VHD disks, raw byte storage

Redundancy — Critical for Exam

Locally Redundant Storage (LRS):

- 3 copies in ONE availability zone in ONE region
- Cheapest option
- Zone outage = potential data loss
- Good for: non-critical, easily reproducible data

Zone-Redundant Storage (ZRS):

- 3 copies across 3 AZs in ONE region
- AZ outage = data survives

- Best balance of cost/resilience for most production
- **Use ZRS for production workloads**

Geo-Redundant Storage (GRS):

- LRS locally + async replica to paired region
- Paired region: e.g., East US ↔ West US
- Secondary is read-only unless you initiate failover
- RPO = ~15 minutes (not zero — you lose last 15 min of writes)

Geo-Zone-Redundant Storage (GZRS):

- ZRS locally + GRS to paired region
- Best of both: AZ resilience locally + geo redundancy
- Max resilience for standard storage

Read-Access Geo-Redundant (RA-GRS / RA-GZRS):

- Secondary region readable always
- Use for: read-only reporting workloads, DR reads
- Only pay for egress on secondary reads

****Exam trap:** GRS/GZRS — the secondary is NOT readable by default. You must explicitly request RA-GRS if you want read access to secondary.**

Storage Account Security

Network access:

- Public endpoints (default) — allow specific IP or VNet
- Private endpoints — private IP in your VNet, no public internet exposure
- Firewall rules — IP allowlist (on-prem access)
- Resource instance rules — allow specific Azure resources

Shared Keys (legacy):

- Two 512-bit access keys
- Full control of everything in the account
- Should be disabled — use Azure AD auth instead

Azure AD authentication (preferred):

- Role-based: Storage Blob Data Reader, Contributor, Owner
- Scope: storage account, container, or blob level
- Supports MSI and service principals

SAS (Shared Access Signatures):

- Delegate access to specific resources
- User delegation SAS (Azure AD auth) — preferred
- Service SAS (account key) — legacy
- Stored Access Policy — associate SAS with policy for revocation
- Account SAS — broad access across all services

Encryption:

- Microsoft-managed keys (default) — AES-256
- Customer-managed keys (CMK) — in Azure Key Vault
- Infrastructure encryption — double encryption at hardware level (compliance requirement)

Azure Blob Storage

Blob Tiers

Tier	Access Tier	Pricing Model	Use When
Hot	Frequent access	Storage cost + low access cost	Active workloads, daily reads
Cool	Infrequent (<30d)	Storage + moderate access	Backup recent data, DR copies
Cold	Rare (<90d)	Storage + higher access	Archival, compliance data
Archive	Very rare (<180d)	Very cheap storage + high retrieval cost	Long-term retention, legal hold

Lifecycle management rules:

```
{
  "rules": [
    {
      "name": "ageOutFiles",
      "enabled": true,
      "prefix": "logs/",
      "expiration": {
        "daysAfterModificationGreaterThan": 90
      },
      "autoTierToHotFromCool": true
    }
  ]
}
```

Standard HTTP access: Blob data accessible via URL: `https://.blob.core.windows.net//`

Immutable Blob Storage (WORM):

- Time-based retention (1 day to forever)
- Legal hold (indefinite until released)
- No delete or overwrite allowed
- Use for: compliance (SEC 17a-4, FINRA), legal discovery

Azure Files

SMB 3.0+ and NFS support — lift-and-shift Windows file shares or Linux NFS workloads.

File Share Tiers

Tier	Protocol	Use When
Transaction Optimized	SMB/NFS	Most workloads — balanced
Hot	SMB only	Frequent access, cloud-native
Cool	SMB only	Infrequent access patterns
Premium	SMB/NFS	High IOPS, low latency (enterprise apps)

Identity-based access:

- AD DS (on-prem or Azure AD DS) — traditional NTFS permissions
- Azure AD Kerberos — native Azure AD integration (no AD required)
- Share-level + file-level permissions

File Sync (Azure File Sync):

- Deploy File Sync agent on Windows Servers on-prem
- Tiering: hot data stays on-prem, cold data moves to Azure Files
- Bandwidth optimization: only sync delta changes
- Cloud-side sharing: users access files via DFS-N or direct share

Exam scenario: "100 remote offices, intermittent connectivity, need local file access + cloud backup" → Azure File Sync with cloud tiering.

Azure Data Lake Storage (ADLS Gen2)

Hierarchical namespace — turns blob storage into a proper file system with POSIX-like permissions.

Why ADLS Gen2 Over Blob?

Feature	Blob	ADLS Gen2
Hierarchical namespace	■ (flat)	■ (directories, files)
POSIX permissions	■ (ACLs at storage level)	■ (filesystem ACLs)
rename/move atomic	■	■
Directory rename cost	O(n)	O(1)
Efficient for Spark/Hadoop	■	■

Security:

- Azure RBAC (storage account level)
- POSIX ACLs (file/folder level) — supports Apache Spark, Databricks
- Hierarchical namespace must be **enabled at creation** — cannot be added later

Use when:

- Data lake architecture (raw → processed → curated)
- Big data analytics (Synapse, Databricks, HDInsight, Spark)
- Data engineering pipelines
- ML training data

Azure SQL Family

SQL Deployment Options — Know the Differences

Service	Scope	Use When
Azure SQL VM	Full VM with SQL installed	100% lift-and-shift, SQL features not in PaaS
SQL Elastic Pool	Shared resources	Many DBs with variable usage, cost pooling
SQL MI (Managed Instance)	Full SQL Server instance	Lift-and-shift with Instance-level features (linked servers, Agent, CLR)
Azure SQL Database	Single database	Cloud-native, most PaaS scenarios

Azure SQL Database — Serverless vs. Provisioned

Provisioned:

- Fixed compute always running
- Predictable billing (compute + storage)
- Auto-pause not available
- Good for: predictable, steady-state workloads

Serverless:

- Auto-pauses when idle (configurable delay, e.g., 1 hour)
- Auto-scales compute based on demand
- Billing per second for compute
- Good for: infrequent, unpredictable, development/test

Hyperscale:

- 100GB to 64TB storage
- Read replicas (up to 4) — scale reads horizontally
- Used by: large production DBs (SaaS, e-commerce)
- Not available for managed instance

SQL Security — Defense in Depth

Network: Private endpoint (no public internet) + firewall rules + VNet integration

Authentication: Azure AD only (preferred) or SQL auth (legacy)

Authorization: Role-based (db_datareader, db_datawriter) + contained DB users

Threat protection: Microsoft Defender for SQL (anomaly detection, vulnerability assessment)

Data encryption: TDE (transparent data encryption) at rest + TLS in transit

Audit: SQL Audit logs to blob storage or Log Analytics

Azure SQL — Geo-Replication

Active geo-replication:

- Up to 4 readable secondaries
- Can be in same region or cross-region
- Read-scale out — route reporting queries to secondary
- Manual failover (no automatic failover for geo-replicas)
- **For automatic failover** → **Auto-failover groups**

Failover groups:

- Group of databases on a SQL Database server
- Automatic replication + failover to paired region
- Listener endpoint — app reconnects automatically after failover

- RTO ~1 hour (for geo-secondary to become primary)
- RPO ~1 hour (max data loss)

Managed Instance geo-replication:

- Same concept but instance-level
- Linked to primary MI in paired region

Azure SQL — Backup

- **Full:** Weekly automatic
- **Differential:** Every 12-24 hours
- **Log backups:** Every 5-10 minutes
- **Point-in-time restore (PITR):** Up to 35 days (LTR up to 10 years)
- **Long-term retention (LTR):** Weekly/monthly backups kept 7 days to 10 years in blob storage

Azure Cosmos DB

Multi-model NoSQL database — globally distributed by default.

API Models — Which to Use When

API	Data Model	Use When
SQL (Core)	Document JSON	Most flexible, any schema, serverless
Cassandra	Wide column	Migrating from Cassandra, existing drivers
MongoDB	Document	Migrating from MongoDB, document patterns
Gremlin	Graph	Social graphs, fraud detection, recommendations
Table	Key-value	Migrating from Azure Table Storage, key-value

Throughput Models

Manual RU/s (reserved):

- Fixed provisioned throughput
- Pay for RU/s regardless of actual usage
- Good for: predictable, steady workloads

Autoscale RU/s:

- Scales between min and max RU/s automatically
- Only pays for what you use (scales down when idle)
- Good for: variable, bursty workloads

Serverless:

- Pay per operation (no RU provisioning)
- No minimum, no idle cost
- **No multi-region** — single region only
- Good for: < 1TB, unpredictable infrequent access

****Exam trap:** "Workload goes from 0 traffic to 10,000 RUs burst" → Autoscale, not serverless. Serverless can't handle burst.**

Consistency Levels

Level	Description	When
Strong	Linearizability	Financial, inventory — absolute consistency
Bounded Staleness	Within K versions or T time	Geospatial, reporting
Session	Own writes readable	User profile, shopping cart
Consistent Prefix	No out-of-order	Analytics
Eventual	Best effort	IoT telemetry, comments, likes

****Exam tip:** Most exams use **Session** (default for SQL API) or **Bounded Staleness**. Strong has latency cost.**

Cosmos DB Global Distribution

- Multi-region writes (conflict resolution strategies: LWW, custom)
- Automatic failover if region goes down
- Manual failover for maintenance
- Conflict resolution: Last Write Wins (LWW) or custom merge stored procedure
- You can add/remove regions at any time — zero downtime

Multi-region vs. Single-region:

- Multi-region: \$\$\$

Azure Data Factory (ADF)

Managed ETL/ELT in the cloud.

Key Components

- **Pipeline:** Orchestrates activities (copy, execute notebook, run lookup)
- **Activity:** Individual operation (copy data, data flow, web hook)
- **Dataset:** Pointer to data (blob, SQL, ADLS, HTTP)
- **Linked Service:** Connection string to data store
- **Integration Runtime:** Compute engine (Azure, on-prem, SSIS)

Integration Runtimes:

IR Type	Use For
Azure IR	Cloud-to-cloud data movement
Self-hosted IR (SHIR)	On-prem data sources, VNet-protected data
Azure-SSIS IR	Lift-and-shift SSIS packages

ADF Copy Activity:

- Source → Sink
- Supports 90+ connectors
- Copy can be parallel (performance tuning)
- Schema mapping and data transformation

Data Flow (ETL visual):

- Spark-based transformation in ADF
- No code — visual data flow designer
- Runs on Azure Databricks cluster (auto-provisioned)

■ DOMAIN 3 — Compute & Application Solutions (25–30%)

Azure Virtual Machines

VM Selection Guide

Series	Category	Use Case
A	Basic/Development	Dev/test, lightweight workloads
B	Burstable	Variable CPU usage, burst when needed
D	General Purpose	Most workloads, balanced CPU/memory
E	Memory Optimized	SQL, SAP, in-memory DBs
F	Compute Optimized	Batch processing, gaming servers
G	GPU	ML training, VDI, rendering
H	HPC	High-performance computing
Ls	Storage Optimized	Big data, Data Lake, NoSQL
M	Memory Large	Large SAP, SQL Server Enterprise

Av2-series: Older, basic tier — avoid for production.

B-series (Burstable):

- Base CPU performance + burst credits
- Use credits when you need more CPU
- Good for: dev/test, websites with traffic spikes, small DBs
- Monitor credit balance — if depleted, VM runs at base speed

M-series:

- Up to 4TB RAM (M416ms_v2 = 4TB, 416 vCPUs)
- SAP HANA certified instances
- Very expensive — use only when required

VM Disks

Disk Type	Max IOPS	Max Throughput	Use
Standard HDD	~500	~60 MB/s	Dev/test, cheap storage
Standard SSD	~2,000	~60 MB/s	Production dev/test workloads
Premium SSD	Up to 20,000	Up to 900 MB/s	Production, business critical
Ultra Disk	Up to 400,000	Up to 4,000 MB/s	Highest performance, SAP HANA, big data

Ultra Disk: Attach as data disk only, not OS. Separate compute + storage billing.

Managed Disks vs. Unmanaged:

- Managed: Azure manages storage accounts, better SLA, Azure Backup integration
- Unmanaged: You manage storage account — NOT recommended

Disk encryption:

- SSE (Storage Service Encryption) — default, Microsoft-managed keys
- CMK via Azure Disk Encryption (ADE) — Windows BitLocker / Linux DM-Crypt
- Encryption at host — VM host server encrypts before data written to disk

VM High Availability

Availability Set:

- 2 or more VMs in same DC, different fault/update domains
- Protects against hardware failure within a single DC
- FD = different rack
- UD = different power/network
- **SLA: 99.95%** (when using 2+ VMs in same availability set)

Availability Zones:

- Physically separate DCs within a region
- 3 zones per region (not all regions have 3)
- Protects against entire DC outage
- VMs in different zones = **99.99% SLA**
- ZRS for managed disks in zone-redundant storage

Proximity Placement Groups:

- Keep VMs physically close to minimize latency
- Used for: SAP, HPC, low-latency cluster workloads
- Anti-affinity: spread VMs across zones

VM Scale Sets (VMSS)

- Auto-scale based on CPU, memory, custom metrics
- Load balancer or Application Gateway in front
- Uniform vs. Flexible orchestration mode
- Health extension for monitoring VM health
- Overprovisioning: launches extra VMs to ensure desired capacity during scaling

When to use VMSS:

- Hundreds of identical VMs
- Auto-scale web apps (stateless workloads)
- Batch processing
- Microservices infrastructure

Azure App Service

PaaS for web apps, API apps, and background jobs. Fully managed, auto-patching, auto-scale.

App Service Plans

Tier	Workers	Use When
Free / Shared	Shared	Dev/test, lightweight apps
Basic (B1/B2/B3)	1-3	Dev/test production, low traffic
Standard (S1/S2/S3)	4+	Production web apps, auto-scale
Premium (P0V3-P5V3)	4+	Enterprise, larger scale, more slots
Isolated (I1-I5)	Dedicated hosts	Highest security, compliance

App Service Environments (ASE):

- **ASE v3:** Single-tenant, deployed into your VNet (I1-I5 tiers)
- ASE v3 is generally available — no ASE v2
- Good for: high security, compliance, needing VNet integration without constraints
- Premium v3 uses Isolated SKU

Deployment:

- ZIP deploy, WAR deploy, Docker container (Web App for Containers)
- Deployment slots (staging/preview/prod)
- Swap with preview (test before production traffic)
- Auto-swap when validated

Authentication: EasyAuth — built-in Entra ID / Facebook / Google / Twitter auth, no code required.

Networking:

- VNet integration (regional) — call VNet resources without exposing to internet
- Hybrid connections — access on-prem resources via relay
- Private endpoint — access app from VNet only (no public internet)

Azure Kubernetes Service (AKS)

Key Concepts

Cluster architecture:

- Control plane (managed by Azure): API server, etcd, scheduler, kubelet
- Node pools: set of VMs running kubelet + container runtime
- System node pool (runs system pods): at least 2 nodes for production
- User node pools: your application workloads

Container runtime: Azure uses containerd by default (not Docker) as of 2024.

Node pools:

- System pool: small, at least 2 nodes, don't run app workloads
- User pools: sized for your app, can be Windows or Linux
- Spot pools: cheap, preemptible nodes (Kubernetes supports taints/tolerations)
- Autoscaler: scale node count based on demand

AKS Networking Models

Kubenet (basic):

- Azure CNI assigns VNet IP to pods
- Simpler, limited pod networking
- Pods get IPs from separate subnet
- VNet integration requires additional config

Azure CNI (overlay):

- Pods get IPs from a logical overlay network
- Pod-to-pod communication stays in overlay
- Outbound to VNet/internet via node subnet
- Better for large clusters, advanced networking

Network Policy:

- Kubernetes Network Policy (Calico) — namespace/label-based rules
- Azure NSG at subnet level as alternative

AKS Security

Feature	What It Does
Azure AD integration	Cluster auth via Entra ID
Node pool RBAC	Role bindings for namespaces
Pod Security Standards	Restrict pod capabilities
Secrets in Key Vault	CSI driver mounts Key Vault as volume
Azure Policy addon	Enforce governance at cluster level
Azure Defender for Containers	Runtime threat detection

Azure Key Vault as volume:

```
volumes:  
- name: secrets  
  csi:  
    driver: secrets-store.csi.k8s.io  
    readOnly: true  
    volumeAttributes:  
      secretProviderClass: "azure-kv"
```

Azure Functions

Serverless compute. You write code, Azure handles infrastructure.

Hosting Plans

Plan	Cold Start	Billing	Use When
Consumption	1-2 seconds	Per execution, 400,000 GB-s free	Infrequent, variable workloads
Premium	< 100ms (warm)	Always warm, vCPU/s memory reserved	Always-on, enterprise
Dedicated (App Service)	N/A	On your App Service plan	Existing plan, long-running

Premium features:

- Always warm instances (no cold start)
- VNet integration
- Unlimited execution duration
- More memory and compute

Durable Functions:

- Stateful workflows in serverless
- Orchestrator functions: define workflow
- Activity functions: individual steps
- Entity functions: durable stateful objects
- Use for: approval workflows, long-running processes

Function App: Container for functions sharing config, deployment slot, scaling.

■ DOMAIN 4 — MIGRATIONS (5–10%)

> ■ This is the heavy section — the user specifically asked for migration depth.

Azure Migrate Overview

Azure Migrate = Hub for migration projects. Discovers, assesses, migrates.

Discovery

- **Agentless discovery:** VM analyzer connects to vCenter, extracts VM metadata without agents

- **Agent-based discovery:** Lightweight agent on VMs for app-level discovery
- **Hyper-V discovery:** Uses Hyper-V provider
- **Physical server:** Bootable USB or agent
- **Warehouse discovery:** Connect to data warehouses

Assessment

Assessment criteria:

- Performance-based sizing (right-size based on actual utilization)
- As-on-premises sizing (same specs, lift-and-shift)
- Azure readiness (suitable / partially suitable / suitable with remediation)
- Estimated Azure cost

Assessment outputs:

- Azure readiness classification
- Monthly cost estimate (compute + storage)
- Suggested VM size
- Network bandwidth requirements

Azure Migrate Appliance

- **VMware/Hyper-V:** Deployed as a VM on-prem
- Runs discovery + assessment agents
- Data sent to Azure Migrate (via internet or private link)
- Appliance requires: 16GB RAM, 6 cores, 80GB disk, internet connectivity

Azure Site Recovery (ASR)

The disaster recovery engine — not just for migrations, but continuous replication.

What ASR Does

- **Continuous replication** of VMs to Azure (or secondary site)
- **Automated failover** when primary site goes down
- **Test failover** (drill without impacting production)
- **Recovery plans** — orchestrate multi-tier app failover order
- **RPO/RTO configurable** based on replication frequency

ASR: Azure-to-Azure (Cross-Region DR)

- Replicate Azure VMs from one region to another
- Fully managed by ASR service
- Enable from Azure Portal in minutes
- Managed disks automatically replicated
- Recovery regions: your choice (usually paired region)

Replication policy:

- RPO threshold (e.g., 1 minute — how much data loss is acceptable)
- Recovery point retention (how many recovery points to keep)
- App-consistent snapshots (VSS for Windows, freeze-script for Linux)

Failover types:

- **Planned failover** — no data loss, planned maintenance
- **Unplanned failover** — disaster scenario, potential data loss
- **Test failover** — drill, no production impact

Recovery Plan:

```
Group 1 (Tier 0): AD + DNS (first)
Group 2 (Tier 1): App servers
Group 3 (Tier 2): Database servers (last)
Group 4: Post-steps (update DNS, runbooks)
```

ASR: On-Premises to Azure

For VMware/Physical:

Deploy **Process Server** (replication appliance) on-prem
 Deploy **Configuration Server** (manages replication)

VMs get **Mobility Agent** installed (push or manual)

Replication: On-prem → Process Server → Azure Storage

Master Target Server in Azure creates the replicated VMs

Unified Agent: Single agent for VMware + physical servers (replaces old separate agents).

For Hyper-V:

- No agent needed — uses Hyper-V replica
- Hyper-V host replication → Azure
- System Center VMM supported (if using SCVMM, use SCVMM integration)

Azure VMware Solution (AVS) — Migration

AVS = VMware SDDC running natively on Azure. You don't manage the VMware stack.

HCX (Hybrid Cloud Extension) — The AVS Migration Tool:

Migration Method	Downtime	Scale	How It Works
Cold Migration	High	Small	VM powered off, NFC copy
HCX vMotion	None	Small	Live migration, serial
Bulk Migration	Minimal	Large	Parallel, source off/dest on
Replication Assisted vMotion (RAV)	None	Large	Parallel, continuous replication
OS Assisted Migration	Conversion time	Any	KVM/Hyper-V to vSphere

HCX Service Mesh:

- Connect on-prem vCenter to AVS private cloud
- Network extension: L2 stretch from on-prem to AVS
- VM IP/MAC preserved during migration
- **Best for lift-and-shift:** Move VMs without re-IP addressing

AVS is ideal when:

- VMware license exhausted / expensive (Broadcom pricing crisis)
- Want Azure benefits but can't re-architect
- Regulatory requirement for VMware
- Migration without redesign

AVS vs. Native Azure:

- AVS = VMware on Azure (lift-and-shift)
- Native Azure = re-architect to Azure services
- AVS is expensive but reduces migration complexity

Azure Database Migration Service (DMS)

Fully managed service for migrating databases to Azure with minimal downtime.

Migration Modes

Offline migration:

- Source DB taken offline at migration start
- All data migrated in one shot
- Simpler but requires downtime
- DMS copies schema + data

Online migration:

- Continuous replication while source stays live
- Minimal downtime (cutover is quick)
- Uses change data capture (CDC)
- More complex, requires more planning

Supported Source → Target Paths

Source	Target
SQL Server (on-prem, Azure VM, AWS RDS)	Azure SQL DB, SQL MI, Azure VM

Source	Target
Oracle	Azure SQL DB (Poly), Oracle on Azure VM
PostgreSQL (on-prem, AWS RDS)	Azure Database for PostgreSQL
MySQL (on-prem, AWS RDS)	Azure Database for MySQL
MongoDB	Azure Cosmos DB (MongoDB API)
SAP ASE (Sybase)	SAP ASE on Azure VM

DMS Workflow

Create DMS instance — choose region, SKU (Standard / Premium)

Create migration project — source type + target type

Connect to source — credentials, connection string

Connect to target — Azure DB connection

Run assessment — Data Migration Assistant (DMA) checks compatibility

Configure migration — select tables, mapping

Cutover — go/no-go based on replication lag

Complete migration — source offline, target live

Data Migration Assistant (DMA)

- Free tool you download locally
- Runs **assessment** before DMS migration
- Reports: compatibility issues, breaking changes, behavior changes
- Example: migrating SQL Server → Azure SQL DB: detects features used that aren't supported in PaaS

Common compatibility blockers:

- `sp_executesql` with parameterized queries (varies)
- Linked servers (SQL MI doesn't support)
- SQL Server Agent jobs (need to recreate in Azure)
- CLR assemblies (not supported in SQL DB, only MI)
- Service Broker (not in SQL DB Standard)

Azure DMS v3 (New Version)

- Single-phase migration (assessment → migrate in one flow)
- Built-in SKU recommendations
- No VM deployment needed (fully managed)
- Better monitoring and troubleshooting

SQL Server to Azure SQL — Migration Paths

Lift-and-Shift: SQL Server on Azure VM

- 100% compatible — runs full SQL Server
- Same tools, same features, same management
- You manage the VM + SQL Server
- **Use when:** need features not in PaaS, complex replication, legacy apps
- Azure Hybrid Benefit: use existing SQL licenses

Modernize: SQL Server → Azure SQL DB / MI

Azure SQL Managed Instance:

- Near 100% SQL Server compatibility
- Instance-level features: linked servers, SQL Agent, CLR, Service Broker
- **Migrate with DMA:** Assessment → Validate → Migrate via DMS
- Network: ExpressRoute or VPN to VNet, private endpoint

Azure SQL Database:

- PaaS — fully managed
- **Does NOT support:** linked servers, Agent jobs, CLR, Service Broker
- **Supports:** SQL auth, contained DB users, elastic pools
- Migration: DMA assessment required first

Migration Assessment Checklist

- DMA assessment completed – 0 critical blockers
- Azure AD auth configured (if migrating from Windows Auth)
- Network connectivity: ExpressRoute or VPN established
- Target SKU sized based on assessment (performance-based)
- Database collation compatible
- Firewall rules migrated (Azure SQL firewall)
- Login/USER mapping updated post-migration
- SQL Agent jobs recreated (if MI)
- Backups configured on new target
- Application connection strings updated
- Monitoring + alerts enabled on new instance

AD / Identity Migration

Migrating to Microsoft Entra ID

Azure AD Connect scenarios:

Greenfield (new Azure AD):

- Cloud-only identities from day one
- Simple, no sync complexity

Brownfield (existing on-prem AD):

- PHS: Simple, sync password hashes
- PTA: No hash storage in cloud
- Federation: Full ADFS trust (complex)

Multi-Forest AD Migration

Scenario: Two or more AD forests → single Entra ID tenant.

Options:

Multiple Azure AD Connect servers — each forest has its own sync

- Each AAD Connect has its own sync scope
- Cloud anchor: objectSID for users, netbiosdomain\username for UPN conflicts
- Cannot have conflicting user principal names

MIM (Microsoft Identity Manager) — complex consolidation

- For very complex multi-forest scenarios
- Advanced identity mapping and de-duplication
- High complexity, high cost

Exchange Hybrid:

- Azure AD Connect with Exchange Hybrid configuration
- Mail routing through Exchange Online
- Mailboxes: on-prem + cloud simultaneously
- Required for seamless Exchange → Exchange Online migration

Azure AD Connect Health

- Monitor sync errors
- Alert on: object not synced, password hash sync failure, health agent down
- Dashboard: UPN suffix, forest count, object statistics
- **Critical for migrations:** Before cutting over, ensure sync is healthy

Application Migration Patterns

Lift-and-Shift (Rehost)

- Move VMs as-is to Azure
- Azure Migrate Server Migration
- Fast, low risk, no code changes
- Not optimized — may overpay for resources

Lift-and-Shift with Optimization

- Migrate, then right-size after migration

- Use Azure Migrate to get performance data post-migration
- Adjust VM sizes, remove unused disks

Refactor (PaaS Shift)

- Move app tier to App Service / AKS
- Database to Azure SQL / Cosmos DB
- Requires testing, possible code changes
- Long-term cost savings and manageability

Re-architect

- Rewrite significant portions for cloud-native
- Microservices on AKS
- Event-driven on Azure Functions
- Expensive, long timeline, maximum benefit

Repurchase (SaaS)

- Move to SaaS (e.g., on-prem CRM → Dynamics 365)
- Eliminating management overhead
- Licensing changes

Azure Migration — Network Considerations

Connectivity Options for Migration

Method	Use When
Site-to-Site VPN	< 1Gbps, single region, low complexity
ExpressRoute	> 1Gbps, multi-region, private connectivity, SLA
ExpressRoute Global Reach	On-prem to on-prem via Azure
Azure VWAN	Large branch-to-cloud, global
Azure Virtual Network Peering	VNet to VNet within/between regions
Private Link / Private Endpoint	Access Azure PaaS without going over internet

DNS Migration

- **Azure Private DNS zones:** Auto-registration of VMs
- **On-prem DNS integration:** Conditional forwarders to Azure
- **Hybrid DNS:** Split-brain — different answers for on-prem vs. cloud
- **Private Resolver:** On-prem can query Azure DNS without VPN

Traffic Routing After Migration

- **Azure Front Door / App Gateway:** Global load balancer + WAF
- **Traffic Manager:** DNS-based routing (failover, performance, priority)
- **Azure Load Balancer:** Layer 4, any port
- **Application Gateway:** Layer 7, HTTP/HTTPS, URL routing, WAF

DOMAIN 5 — Networking (Cross-Cutting)

VNet & Subnet Design

VNet: Private address space in Azure (RFC 1918).

Subnets: Segmentation within VNet.

RFC 1918 Block	Range
/16	10.0.0.0 – 10.255.255.255
/12	172.16.0.0 – 172.31.255.255
/8	192.168.0.0 – 192.168.255.255

Best practices:

- Use /24 or /26 per subnet (don't go too small)
- Reserve address space for growth
- Don't use entire /8 for one VNet

Subnets:

- Azure reserves 5 addresses per subnet (first 4 + last 1)
- GatewaySubnet: must be /27 or larger for VPN/ER gateways

Network Security Groups (NSG)

- Layer 3-4 firewall at subnet or NIC level
- Priority: 100–4096 (lower = higher priority)
- Stateful: return traffic auto-allowed
- Default rules (cannot be deleted):
- Allow VNet inbound/outbound
- Allow Azure Load Balancer
- Deny all else

Azure Private Link

Private Endpoint: Brings Azure PaaS into your VNet with a private IP.

- Storage: *.privatelink.blob.core.windows.net
- SQL: *.privatelink.database.windows.net
- No public IP on the PaaS resource
- Traffic never leaves Microsoft network
- Zone-level DNS resolution for private connectivity

Private Link vs. Service Endpoint:

	Private Link	Service Endpoint
Traffic	Through Microsoft backbone	Through Azure backbone
Access	Any VNet + on-prem (PEAM)	VNet subnets only
MS backbon	■ Always	■ Within Azure
On-prem access	■ Via VPN/ER	■ VNet only

ExpressRoute vs. VPN

	ExpressRoute	Site-to-Site VPN
Connectivity	Private, dedicated circuit	Over internet
Bandwidth	Up to 100 Gbps	Up to 1.25 Gbps per tunnel
SLA	99.95% (dual-carrier)	99.9%
Latency	Lower, deterministic	Variable
Encryption	MACsec (layer 2)	IPsec (tunnel)
Cost	Monthly + egress	Metered
Use for	Enterprise production, hybrid	Smaller, dev/test

ExpressRoute Direct: Connect at Microsoft edge directly, up to 100Gbps.

■ DOMAIN 6 — Business Continuity & Disaster Recovery (10–15%)**Recovery Objectives**

Metric	What It Measures
RTO (Recovery Time Objective)	How long until service is back online

Metric	What It Measures
RPO (Recovery Point Objective)	How much data loss is acceptable
RPO = 0	Zero data loss — synchronous replication
RPO = 1 hour	Accept losing up to 1 hour of data

Azure Backup

What it backs up:

Workload	Agent / Method
Azure VMs (Windows/Linux)	VM extension (no agent needed)
On-prem VMs	MARS agent
SQL Server in VM	SQL VSS Writer + MARS agent
Files/folders	MARS agent
SAP HANA in VM	HANA backup extension
Blobs	Immutable + lifecycle policies
Azure Files	Built-in backup (share-level)

Backup policies:

- Daily / weekly / monthly / yearly schedules
- Retention: immediate (snapshot) + long-term (LTR)
- Instant restore: snapshot mounted directly, no copy needed

Azure Backup for SQL in Azure VM:

- Full, differential, log backups
- PITR to any point within retention
- Backup to local disk + vault (geo-redundant)

Azure Site Recovery — DR Design

DR architecture decision tree:

```

Is the workload mission-critical?
■■■ YES → Multi-region DR with ASR
■ - RPO: 1-60 min (configurable)
■ - RTO: < 1 hour (with recovery plan)
■ - Test failover quarterly
■
■■■ PARTIAL → Single-region with Availability Zones
■ - AZ outage protection
■ - Zone-redundant storage
■ - RTO depends on restart/reprovision
■
■■■ NON-CRITICAL → Backup only
- Azure Backup
- RTO: time to restore from vault

```

DR for multi-tier apps:

- Use **Recovery Plans** to orchestrate tier-by-tier failover
- Dependencies must be mapped
- Network: Azure Bastion, DNS updates, connection strings
- Test: Run DR drill at least quarterly

High Availability — Azure-native

Zonal services (deploy to specific AZ):

- VMs with Availability Zone (zonal)
- Managed disks with ZRS
- Public IP Standard with zone■■■
- Load Balancer Standard with zone■■■

Zone-redundant services:

- Azure SQL (zone-redundant HA)
- Cosmos DB (zone redundancy option)
- ADLS Gen2 (ZRS)
- Event Hubs (zone redundancy)

Availability Set vs. Zone:

	Availability Set	Availability Zone
Scope	Single DC	Multiple DCs in region
Protects	Hardware fault	Entire AZ outage
Cost	Included	Per-zone charges
Complexity	Lower	Higher

COST OPTIMIZATION

Right-Sizing

- Azure Migrate assessment gives sizing recommendations
- Azure Advisor: Right-size underutilized VMs
- Azure Reserved Instances: 1 or 3 year commitment = 72% savings vs. pay-as-you-go
- Azure Hybrid Benefit: Use existing Windows Server / SQL Server licenses

Compute Savings Options

Option	Savings	Use When
Reserved Instances (1yr/3yr)	Up to 72%	Steady, predictable workloads
Savings Plans (1yr/3yr)	Up to 60%	Flexible, any compute
Spot VMs	Up to 90%	Interruption-tolerant batch, dev/test
Azure Hybrid Benefit	100% of license cost	Existing Windows/SQL licenses
Dev/Test pricing	Up to 50%	Development/sandbox subscriptions

Storage Cost Optimization

- Use Cool/Cold/Archive tiers for infrequently accessed data
- Lifecycle policies: auto-transition from Hot → Cool → Archive
- ZRS only where needed (use LRS for non-critical)
- Soft delete + immutability for compliance at minimum extra cost
- Azure Files snapshots for cheap backup

AZURE WELL-ARCHITECTED FRAMEWORK

Pillar	Key Focus
Cost	Right-size, reserve, spot, monitor spend
Resiliency	HA/DR design, SLA awareness, automatic recovery
Security	Defense in depth, Zero Trust, least privilege
Performance	Auto-scale, caching, CDN, async patterns
Operations	Monitoring, alerting, IaC, automation

QUICK REFERENCE — Decision Trees

Which Database?

Need relational?
■■■ SQL Server compatibility (full)?
■ ■■■ Need linked servers / Agent / CLR?
■ ■■■ YES → SQL Managed Instance
■ ■■■ NO → Azure SQL Database
■ ■■■ Lift-and-shift existing SQL Server VM?
■ ■■■ YES → Azure VM with SQL
■■■ NoSQL / Flexible schema?
■■■ Global distribution?
■ ■■■ YES → Cosmos DB
■ ■■■ NO → Azure Table Storage
■■■ Analytics / Data Lake?
■■■ YES → ADLS Gen2 + Synapse

Which Migration Path?

Is it a VM?
■■■ Lift-and-shift → Azure Migrate (ASR)
■■■ Replatform → Azure Migrate + right-size
■■■ Refactor → App Service / AKS
Is it a database?
■■■ SQL Server? → DMA → DMS → Azure SQL / MI
■■■ Oracle? → DMS → Azure VM with Oracle
■■■ PostgreSQL/MySQL? → DMS → Azure Database for PostgreSQL/MySQL
Is it VMware?
■■■ AVS compatible? → Azure VMware Solution + HCX
■■■ Can re-architect? → Native Azure PaaS

Which Identity?

On-prem AD exists?
■■■ YES → Azure AD Connect (PHS / PTA / Federation)
■ ■■■ Simple sync, cloud fallback → PHS
■ ■■■ No hash in cloud, MFA on-prem → PTA
■ ■■■ Complex auth, smart card → Federation
■■■ NO → Cloud-only Entra ID
Need SSO to SaaS?
■■■ YES → Entra ID Application Proxy / SAML
■■■ NO → Direct auth

■ Exam Day Tips

Read the scenario first — identify the business requirement before looking at options

RTO/RPO questions — know which service supports which RPO

"Least disruptive" → prefer lift-and-shift over re-architect

"Most cost-effective" → right-sizing + reserved capacity

"Zero data loss" → synchronous replication (Availability Zones)

"Global distribution" → Cosmos DB, Front Door, Traffic Manager

"Hybrid identity" → Azure AD Connect + Entra ID

"VMware on Azure" → Azure VMware Solution (HCX for migration)

"Multi-region DR" → ASR with failover groups or Cosmos DB multi-region

"Serverless" → Azure Functions, Logic Apps, Azure SQL Serverless

Study hard. Pass the exam. ■

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